

Adapted from 2024 VJC Prelims Paper 2

Answer **three** questions in total.

Section A

One or two of your three chosen questions must be from this section.

- 1** In 2022, the Minister for Finance announced that the Goods and Services Tax (GST) rate will be raised in two steps, from 7 per cent to 8 per cent on 1 Jan 2023 and from 8 per cent to 9 per cent on 1 Jan 2024. The revenue from the increase in GST will go towards meeting our medium-term needs including supporting our healthcare expenditure, and to take care of our senior citizens.

Source: www.gov.sg, 1 Jan 2024

- (a) Using relevant elasticity concepts, explain how an increase in income levels and the GST hike can each affect consumers' expenditure on healthcare services. [10]
- (b) Discuss whether the provision of healthcare subsidies or the imposition of a price cap on healthcare services is a better measure to help senior citizens cope with the rising cost of healthcare in Singapore. [15]
- 2** In February 2024, Ritual, a boutique gym brand, abruptly shut all four outlets in Singapore permanently. The company cited COVID-19 as a big factor impacting their business, on top of other challenges like rising rental costs and changing workout fads. Locally owned boutique gyms are hit harder by competition compared to larger gym chains like Fitness First, which typically have more capital and a more attractive brand image.
- (a) Explain why Ritual exited the market in response to the challenges in the gym industry. [10]
- (b) Discuss the factors that a gym would consider when deciding which strategies to adopt to remain in the market. [15]
- 3** Excessive tourism often leads to significant water and air pollution, as increased demand for resources and services strains local environments, leading to inefficiencies and long-term ecological damage.
- (a) Explain how a rational consumer decides whether to travel abroad for leisure, and how such decision-making may lead to an inefficient allocation of resources. [10]
- (b) Discuss the measures a government can implement to improve resource allocation in the market for tourism services. [15]

Section B

One or two of your three chosen questions must be from this section.

- 4** Governments seek to attain various macroeconomic goals, one of which is inclusive and sustainable economic growth, in order to raise the living standards of its citizens.
- (a) Explain why governments aim to achieve inclusive and sustainable economic growth. [10]
- (b) Discuss the extent to which policies to achieve inclusive economic growth would cause difficulties for an economy. [15]
- 5** In early 2022, Sri Lankans started experiencing power cuts and shortages of basics such as fuel. The rate of inflation rose to 50 per cent while unemployment rate remains high. Currently, Sri Lanka owes about \$7 billion to China and around \$1 billion to India.

Adapted from BBC, 29 March 2023

- (a) Explain two possible causes of inflation. [10]
- (b) Discuss the extent to which the Sri Lankan government should focus on reducing its inflation rate over reducing its unemployment rate. [15]
- 6** Deglobalisation is the movement towards a less connected world, characterised by powerful nation states, local solutions, and border controls, rather than global institutions, treaties, and free movement.
- (a) Explain why a country like Singapore exports and imports both different and similar types of goods and services when it trades with other countries like China. [10]
- (b) With the trend of rising deglobalisation, discuss whether Singapore should also deglobalise. [15]

Adapted from 2024 VJC Prelims Paper 2 Suggested Answers

- 1 In 2022, the Minister for Finance announced that the Goods and Services Tax (GST) rate will be raised in two steps, from 7 per cent to 8 per cent on 1 Jan 2023 and from 8 per cent to 9 per cent on 1 Jan 2024. The revenue from the increase in GST will go towards meeting our medium-term needs including supporting our healthcare expenditure, and to take care of our senior citizens.

Source: www.gov.sg, 1 Jan 2024

- (a) Using relevant elasticity concepts, explain how an increase in income levels and the GST hike can each affect consumers' expenditure on healthcare services. [10]
- (b) Discuss whether the provision of healthcare subsidies or the imposition of a price cap on healthcare services is a better measure to help senior citizens cope with the rising cost of healthcare in Singapore. [15]

Part (a) Question Interpretation

Command word/phrase	Explain how	Make clear the reasons and explain the theory behind the question. Cause and effect relationships must be explained thoroughly
Content	<i>Using relevant elasticity concepts</i>	
	<i>Increase in income levels</i>	Relevant elasticity concept of YED
	<i>GST hikes</i>	Relevant elasticity concept of PED as GST hikes will affect the supply
	<i>Consumers' expenditure</i>	Consumers' expenditure = price x quantity
Context	<i>Healthcare services</i>	Apply analysis to the context of healthcare service in Singapore

This question requires an explanation of how a rise in income can lead to an increase in the expenditure for healthcare services by applying income elasticity concept – normal good and an explanation of how a GST hike can lead to an increase in the expenditure for healthcare services by applying price elasticity of demand concept.

Introduction

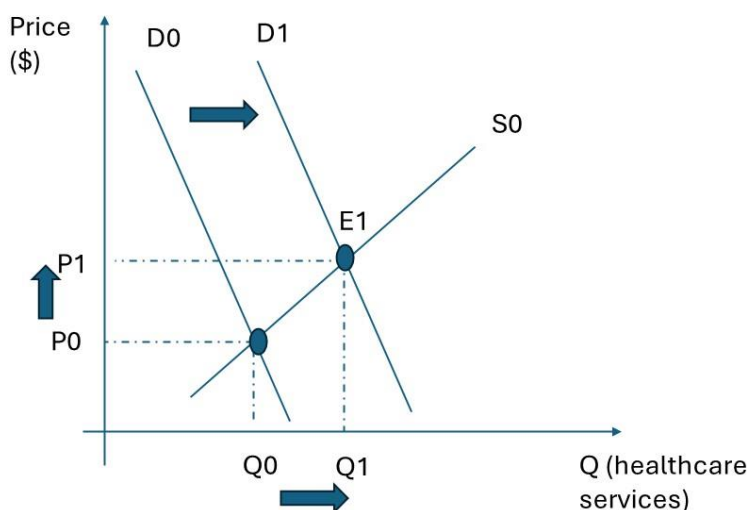
- The rise in income levels as well as GST hike will impact the market for healthcare services in Singapore.
- Using demand and supply analysis, as well as the concepts of income elasticity of demand (YED) and price elasticity of demand (PED), this essay will explain the impact of the two events on the total expenditure on healthcare services by consumers.

Body

Key Argument 1: Explain how a rise in income can lead to an increase in the expenditure for healthcare services (apply income elasticity concept – normal good).

- A rise in income will lead to an increase in consumers' expenditure on healthcare services.
- YED is the responsiveness of demand for a good or service to a change in income level, ceteris paribus.
 - When YED is positive, a rise in income results in a rise in the ability of households to consume more normal goods and services, which include luxury goods ($YED > 1$) and necessities ($0 < YED < 1$).
- Since healthcare services are considered normal services due to their necessity in people's lives (i.e. regular health check-ups to ensure people remain healthy, or doctor visits due to illness), this rise in income would cause the demand for healthcare services to increase.
 - This is illustrated by a rightward shift in the demand curve in Figure 1a below.

Figure 1a. Rise in income on expenditure for healthcare services

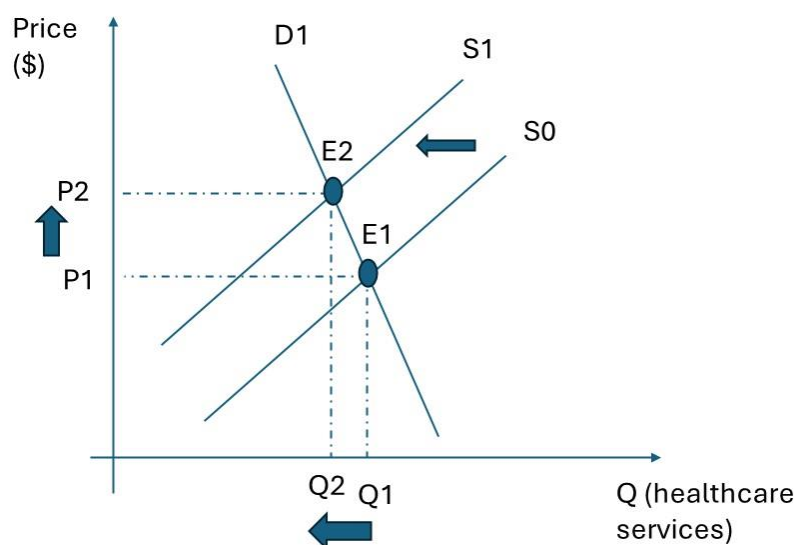


- Holding supply curve constant, the rightward shift in the demand curve from D0 to D1 would lead to a shortage at the initial price, P0.
 - This shortage will lead to an upward pressure on price. As price rises, quantity demanded would fall, as shown by a movement along the new demand curve, D1.
 - At the same time, quantity supplied would also rise as the price rises, as shown by a movement along the original supply curve, S0.
 - The process stops when the new demand curve intersects the supply curve at E1, resulting in a new equilibrium.
- Since expenditure refers to the total amount spent on the good, the rise in equilibrium price and quantity from $(P0 \times Q0)$ to $(P1 \times Q1)$, will result in an increase in total expenditure on healthcare services.

Key Argument 2: Explain how a GST hike can lead to an increase in the expenditure for healthcare services (apply price elasticity of demand concept).

- **A GST hike would lead to an increase in the expenditure for healthcare services.**
- An increase in GST, which is an indirect tax, results in an increase in the marginal cost of production of firms.
 - When this happens, firms will increase the price by the amount of tax at each quantity supplied to maintain profitability.
 - This is illustrated by a leftward shift in the supply curve from S_0 to S_1 in Figure 1b below.
- Holding demand curve D_1 constant, the leftward shift in the supply curve from S_0 to S_1 would lead to a shortage at the initial price, P_1 .
 - This shortage will lead to an upward pressure on price. As price rises, quantity supplied would rise, as shown by a movement along the new supply curve, S_1 .
 - At the same time, quantity demanded would fall as prices rise, as shown by a movement along the demand curve, D_1 .
 - The process stops when the new supply curve intersects the demand curve at E_2 , resulting in a new equilibrium.
- The concept of PED can be used to determine the effect of the GST hike on consumers' expenditure on healthcare services.
- PED measures the responsiveness of quantity demanded to a change in the price of healthcare services, ceteris paribus.
- Since there are no close substitutes for healthcare services and the degree of necessity is high (especially for certain essential healthcare services), the demand for healthcare services is likely to be price inelastic i.e. $|PED| < 1$.
 - Hence, the rise in price because of the GST will lead to a less than proportionate fall in quantity demanded.
- Since the rise in expenditure due to a rise in price is greater than the fall in expenditure due to a fall in quantity demanded, total expenditure on healthcare services will rise because of the GST.

Figure 1b. Hike in GST on expenditure for healthcare services



Conclusion

- As such, the rise in income levels and GST hike would both lead to an increase in consumers' expenditure on healthcare services.

Mark Scheme

Level	Knowledge, Application/Understanding and Analysis	Marks
L3	For a well developed answer that has: Good scope – explains the effects of the two events on consumers' expenditure on healthcare services using the concepts of YED and PED; and Good rigour – supporting demand and supply analysis with accurate and well-labelled diagram(s); and Good application to the context of market for healthcare services	8 – 10
L2	For an answer that: Lacks scope – only explains how one event or incomplete use of elasticity concepts in the explanation on consumers' expenditure on healthcare services. Lacks rigour – there are gaps in using demand and supply analysis and/or explanation is not supported by accurate and well-labelled diagrams(s) where relevant; and/or Lacks application to the context of market for healthcare services.	5 – 7
L1	For an undeveloped answer that: - Is descriptive, lacking in application of economic theory; and/or - Contains serious and pervasive conceptual errors; and/or - Is largely irrelevant.	1 – 4

1 In 2022, the Minister for Finance announced that the Goods and Services Tax (GST) rate will be raised in two steps, from 7 per cent to 8 per cent on 1 Jan 2023 and from 8 per cent to 9 per cent on 1 Jan 2024. The revenue from the increase in GST will go towards meeting our medium-term needs including supporting our healthcare expenditure, and to take care of our senior citizens.

Source: www.gov.sg, 1 Jan 2024

- (b) Discuss whether the provision of healthcare subsidies or the imposition of a price cap on healthcare services is a better measure to help senior citizens cope with the rising cost of healthcare in Singapore. [15]

Part b Question Interpretation

Command word/phrase	Discuss whether...	To examine and present the different aspects of a problem or subject, and give a judgement
Content	<i>Provision of healthcare subsidies</i>	Policy 1: explain how subsidies work & its limitation or unintended consequences
	<i>Imposition of price cap on healthcare services</i>	Policy 2: explain how maximum price/price ceiling work and its limitation or unintended consequences
	<i>Senior citizen to cope with rising cost of healthcare</i>	Objective: reduce the price of healthcare
Context	<i>Singapore</i>	Context of Singapore

A relevant answer requires a balanced analysis (workings and limitation/unintended consequences) of the 2 stated policies that can be used to address the issue of rising healthcare costs borne by the senior citizens in Singapore. A judgement of which is a better measure with substantiation is required.

Introduction

- Given the rising cost of healthcare in Singapore, there is scope for the government intervene to help ensure healthcare costs are affordable, especially for senior citizens, who form an increasing proportion of Singapore's population. This essay will discuss the policies of healthcare subsidies and the imposition of a price ceiling on healthcare services, as well as consider which is likely to be better at helping senior citizens cope with these rising healthcare costs.

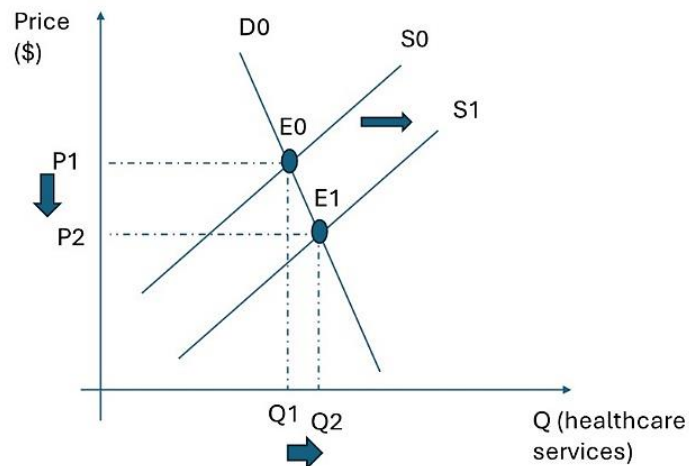
Body

Key Argument 1: Explain how the provision of subsidy (reduce price of healthcare) can help the seniors to cope with rising healthcare costs in Singapore & its limitation/unintended consequence

- The provision of subsidy will lead to a reduction in the price of healthcare to help the seniors to cope with rising healthcare costs in Singapore.**
- A subsidy will lower the marginal cost of producing healthcare services for firms in the market.
 - This means that healthcare providers will be more willing and able to offer healthcare services at each price level, leading to a rise in supply of healthcare services, which is illustrated by a rightward shift in the supply curve from S₀ to S₁.

- This creates a surplus at the initial price, P_1 . This surplus will cause downward pressure on the price.
- As price falls, quantity demanded will rise as illustrated in Figure 2 below by a movement along the demand curve while quantity supplied will fall, as illustrated by a movement along the new supply curve, S_1 .
- The new equilibrium is achieved at point E_1 when the demand curve, D_0 intersects the supply curve, S_1 .

Figure 2. Subsidy provided for healthcare services



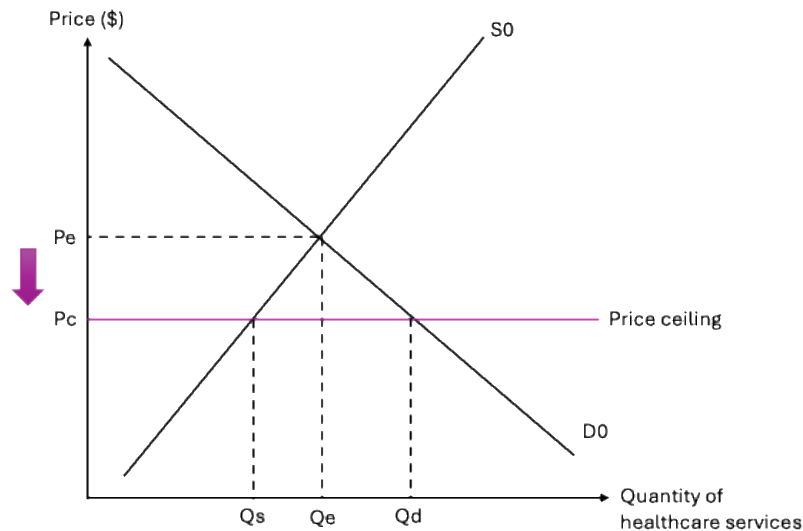
- At the new equilibrium, seniors can now enjoy more healthcare services at a lower price, P_2 , and equilibrium quantity increases to Q_2 .
- This helps senior citizens cope with the rising healthcare costs in Singapore by lowering its price and also increasing the amount of healthcare services offered.

Limitation/unintended consequences:

- The provision of subsidies will put a strain on the government's budget. There is an opportunity cost incurred from the government choosing to spend on subsidising healthcare – in the case of Singapore, this would likely be the net benefits to society that could have been gained had the government spent on education or skills upgrading instead (the next best alternative foregone).
 - Even though Singapore still has large fiscal reserves, if the subsidies were to be provided in the long-term, coupled with our increasingly ageing population, the policy could end up becoming fiscally unsustainable.
 - This is especially so since the demand for healthcare services is price inelastic (as explained in part (a)) – as such, a sizeable subsidy is required to significantly increase the consumption of healthcare by senior citizens to the level desired by the government.

Key Argument 2: Explain how the imposition of a price ceiling can help the senior citizens in Singapore to cope with rising healthcare costs in Singapore & its limitation/unintended consequences

- **The implementation of a price ceiling may be a better measure to help the seniors in Singapore to cope with rising healthcare costs in Singapore.**
- A price ceiling, set below the market equilibrium, is a legal maximum limit on the price of a good or service imposed by the government to protect consumers by ensuring that prices do not become prohibitively expensive.

Figure 3. A price ceiling for healthcare services

- As shown in Figure 3 above, by setting a maximum price, P_c , the quantity demanded would increase from Q_e to Q_d while the quantity supplied falls from Q_e to Q_s .
 - When the price falls to P_c (now the highest price that healthcare providers can legally charge for healthcare services), consumers are now more willing and able to consume more healthcare services.
 - However, producers are now less willing and able to produce the same amount of healthcare services as before, at this lower price.
 - Hence, there will be a persistent shortage of $Q_s Q_d$ units of healthcare services.
- Consequently, an alternative allocation mechanism for healthcare services is required – this could be in the form of providing priority access to healthcare for senior citizens, or via a first-come-first-served basis (i.e. queuing).
- Notwithstanding, the price ceiling has resulted in a lower price for healthcare services that allows seniors to cope with its rising costs.

Limitation/unintended consequences:

- The implementation of a price ceiling will lead to welfare loss as it disrupts the efficient allocation of resources in the market, resulting in a persistent shortage.
- For healthcare services to be offered at a lower price at Q_d , the government will need to ensure that producers can meet the shortage of $Q_s Q_d$ through other forms of subsidies or grants.
 - Otherwise, the fall in quantity supplied of healthcare services from Q_e to Q_s would defeat the purpose of the policy – even though healthcare services have been made more affordable to senior citizens (at a lower price P_c), there is a lower quantity of healthcare services offered.

Evaluative Conclusion (a reasoned stand that follows from the above analysis + 1 well-developed ATMS evaluative angle would suffice):

- [Stand]** The provision of healthcare subsidy is a better measure to help the senior citizens in Singapore cope with rising healthcare costs in Singapore because it is easier to implement and is likely to lead to lower welfare loss to society.
- [Situation]** Having a subsidy would allow all senior citizens to benefit from lower price of healthcare services in general. In addition, the provision of a subsidy does not have the issue of shortage for any of the healthcare services. This is especially so when Singapore is facing a rapidly ageing population.

- **[Magnitude]** Imposing price ceiling on the provision of healthcare services leads to welfare loss because of the persistent shortage that is created. As such, the government will still need to provide sufficient subsidy in order for such essential services to be provided.
- **[Alternative]** Alternatively, grants (transfer payments) can be given to senior citizens who are less able to afford certain expensive services such as surgery based on their housing status and insurance coverage. With more targeted assistance and spending by the government, the likelihood of overconsumption and lack of perfect knowledge on the amount of subsidy can be better prevented.

Mark Scheme

Level	Knowledge, Application/Understanding and Analysis	Marks
L3	For a well developed answer that has: • Good scope – explains how the provision of healthcare subsidies and the imposition of a price cap on healthcare services can help senior citizens cope with rising healthcare costs & their limitations/unintended consequences • Good rigour – supporting demand and supply analysis with accurate and well-labelled diagram(s); and • Good application to the context of market for healthcare services	8 – 10
L2	For an answer that: • Lacks scope – only explains how one of the above policies and its limitation/unintended consequences and/or • Lacks rigour – there are gaps in using demand and supply analysis and/or explanation is not supported by accurate and well-labelled diagrams(s) where relevant; and/or • Lacks application to the context of market for healthcare services.	5 – 7
L1	For an undeveloped answer that: • Is descriptive, lacking in application of economic theory; and/or • Contains serious and pervasive conceptual errors; and/or - Is largely irrelevant.	1 – 4
Evaluation		
E3	Well-developed evaluative judgements across both requirements are and backed by sound economic analysis. A summative conclusion that answers the question and takes a stand is provided.	5
E2	Some evaluative comments and attempts at making judgements that are explained using relevant economic analysis.	3 – 4
E1	An evaluative statement with some attempt at economic analysis.	1 – 2

2 In February 2024, Ritual, a boutique gym brand, abruptly shut all four outlets in Singapore permanently. The company cited COVID-19 as a big factor impacting their business, on top of other challenges like rising rental costs and changing workout fads. Locally owned boutique gyms are hit harder by competition compared to larger gym chains like Fitness First, which typically have more capital and a more attractive brand image.

- (a) Explain why Ritual exited the market in response to the challenges in the gym industry. [10]
- (b) Discuss the factors that a gym would consider when deciding which strategies to adopt to remain in the market. [15]

Part (a) Question Interpretation

Command word/phrase	Explain why	To utilise economic reasons to explain why firm exited the industry/market
Content	Challenges faced in the gym industry on firm's decision to exit the industry	Rising rental cost & changing workout fads have affected the cost, revenue and hence profits of firm. Exit market/industry → subnormal profits in the long run
Context	Gym industry	Application to context of gym industry

Students are expected to consider the challenges faced by Ritual in the gym industry and explain the effects of these challenges on the costs, revenue and hence profits made by Ritual. Students need to bring in the exit rule to explain why Ritual decided to leave the market (in the long run).

Introduction

- In the long run, all costs are variable and rational firms will only remain in the market if they can cover all their costs of production (i.e. earn at least normal profits). If a firm ends up making subnormal profits in the long run, it should exit the industry for good.
- From the preamble, the gym industry has been facing myriad challenges that have negatively affected both the revenue and costs of gyms.
- Since profits = total revenue (TR) – total costs (TC), the fall in TR and rise in TC might have led to gyms like Ritual earning subnormal profits in the long run, resulting in it exiting the market.

Body

Key Argument 1: Explain the challenges in the gym industry and their impact on the cost and revenue of firms in the gym industry, such as Ritual.

- COVID-19 as well as rising rental costs led to a rise in cost of production for Ritual.**
- The COVID-19 pandemic resulted in labour shortages and supply chain disruptions that might have limited the ability of gyms to obtain factor inputs such as labour and gym cleaning/sanitisation equipment.
- This led to a rise in both the marginal cost (MC) and average cost (AC) of production for gyms, including Ritual.
 - Furthermore, the rise in rental costs, which is a fixed cost in the short run, would have further increased Ritual's AC, as Ritual needs land space in order to operate its four gym outlets.

- As such, the rise in costs due to the above challenges is illustrated as upward shifts of the MC and AC curves from MC and AC to MC' and AC' respectively.
- Note: It is ok if students only explain one example of a cost factor, but the factor must be well-explained. For students who focus on rental costs, only the AC shifts upwards (in the SR).*
- COVID-19 and changing workout fads led to a fall in demand for Ritual's gym services.**
- During the COVID-19 pandemic, gyms operated on reduced capacity (or were not even allowed to open during the Circuit Breaker), in order to abide by Safe Management Measures.
- Consumers might also have preferred to workout outdoors or at home to avoid contracting COVID-19.
- As a result, there was a fall in the willingness of consumers to go to the gym, leading to reduced demand for Ritual's gym services.
 - Furthermore, changing workout fads might have led some individuals to prefer exercising in larger chain gyms like Fitness First rather than smaller boutique gyms like Ritual, perhaps due to the greater freedom and variety in equipment use in larger gyms.
 - This change in tastes and preferences away from boutique gyms would have also resulted in a fall in the demand for Ritual's gym services.
 - The fall in demand is illustrated as a leftward shift of its DD=AR and MR curves from DD=AR and MR to DD'=AR' and MR' respectively.

Key Argument 2: Explain how the rise in costs and fall in revenue led to subnormal profits for Ritual, hence causing them to exit the market in the long run.

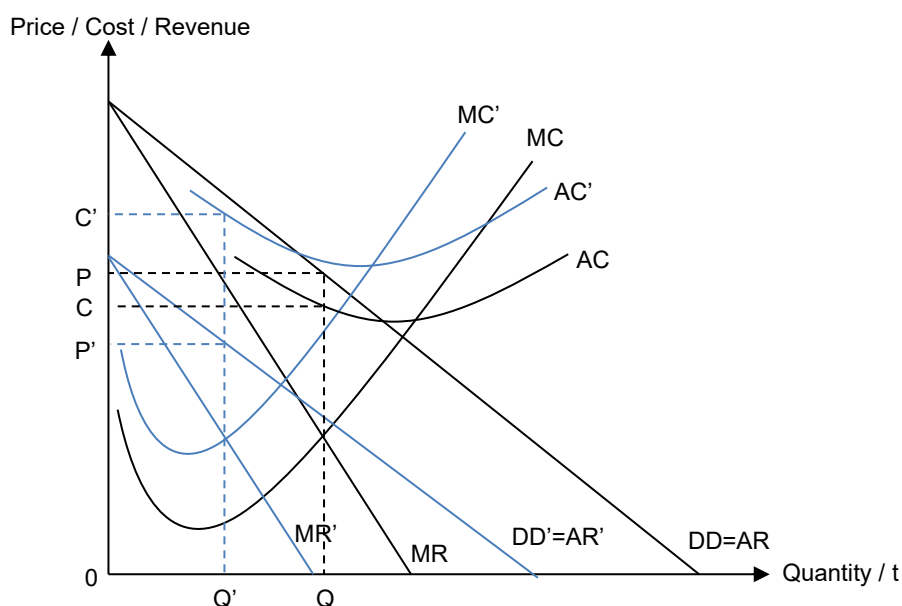


Figure 1: Impact of challenges in the gym industry on Ritual's profits

- The rise in costs and fall in revenue led to Ritual earning subnormal profits.**
- Assuming Ritual's objective is to maximise profits, it was initially producing at output level Q, where $MC=MR$, and priced its gym services at price P (corresponding to the demand curve $DD=AR$). It earned supernormal profits of $[(P-C) \times Q]$.
- However, as a result of the rise in costs and fall in revenue, MC and AC shifted upwards to MC' and AC' respectively, and DD=AR and MR fell to DD'=AR' and MR' as explained above.
 - Ritual's new profit-maximising output level fell to Q', and price rose to P'.

- This resulted in Ritual now earning subnormal profits of $[(C' - P') \times Q']$.
- In the long run, the subnormal profits earned led Ritual to exit the market as it was no longer profitable to remain in the industry.
- As such, the various challenges in the industry led Ritual to exit the market due to the subnormal profits made in the long run.

Conclusion

- In conclusion, Ritual exited the market as the challenges it faced resulted in the gym earning subnormal profits in the long run.

Mark Scheme

Level	Knowledge, Application/Understanding and Analysis	Marks
L3	For a well developed answer that has: • Good scope – explains the impact of challenges in the gym industry on Ritual's costs and revenue, and how it led to Ritual earning subnormal profits in the long run (i.e. exit rule).; and • Good rigour – supporting firm's analysis with accurate and well-labelled diagram(s); and • Good application to the context of gym industry	8 – 10
L2	For an answer that: • Lacks scope – only explains how one part of the question; and/or • Lacks rigour – there are gaps in using firm's analysis and/or explanation is not supported by accurate and well-labelled diagrams(s) where relevant; and/or • Lacks application to the context of gym industry	5 – 7
L1	For an undeveloped answer that: • Is descriptive, lacking in application of economic theory; and/or • Contains serious and pervasive conceptual errors; and/or • Is largely irrelevant.	1 – 4

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(b) Discuss the factors that a gym would consider when deciding which strategies to adopt to remain in the market. [15]

Part (b) Question Interpretation

Command word/phrase	Discuss the factors ... a gym should consider	Explain the factors that firm should consider and come to a substantiated judgment. The decision-making framework can also be used. Consider (1) benefits, (2a) unintended consequence/cost & (2b) constraint/limitation
Content	To remain in the market	Remain in the industry
Context	Gym market	To apply analysis to the context.

Students are expected to elicit at least two factors that a gym would consider when deciding which strategies to adopt to remain in the gym industry. There are many such factors, of varying importance, including the market structure the gym belongs to, the level of profits earned, the firm's objective(s), as well as external conditions (e.g. state of economy, changes in consumer preferences). For evaluation, students need to weigh between the factors to decide which is the most important consideration for a gym when deciding which strategies to adopt to remain in the market.

Alternatively, students may also opt to explain the factors as "benefits" and "costs" (or constraints). This approach is also acceptable:

- *Benefits of strategies: Explain one or two strategies and possible limitations that might hinder the effectiveness of the strategies.*
- *Cost of strategies / constraints: Ideally, link to ability to secure funding*

Introduction

- As explained in (a), various challenges in the gym industry have led to some gyms (e.g. Ritual) exiting the market. In response to these challenges, firms remaining in the industry may decide to implement strategies to remain in the market, by raising revenue or lowering their costs of production.
- However, the strategy or strategies a particular gym would adopt depends on various factors.

Body

Key Argument 1: Explain one factor that a gym would consider when deciding which strategies to adopt to remain in the market.

- **One factor that a gym would consider when deciding which strategies to adopt to remain in the market would be its market structure.**
- Gyms generally fall within the oligopolistic market structure, due to the significant barriers to entry (BTE) faced, e.g. huge cost of capital needed to set up and equip a gym or chain of gyms.
 - As such, gyms operate on a large scale and reap significant internal economies of scale (iEOS), and there are few of such large firms in the industry (e.g. Fitness First, True Fitness, Pure Fitness).

- There is thus mutual interdependence between gyms, in that each firm takes the actions and reactions of their rivals' strategies into account when making their own production and marketing decisions.
- Since gyms are oligopolistic in nature, they tend to avoid price strategies, as they result in lose-lose situations – for example, if Fitness First were to lower their prices in a bid to increase revenue and hence profits, rival firms such as True Fitness would like also lower their prices similarly to avoid losing market share, which may result in both firms lowering prices and hence profits but without enjoying any eventual increase in market share.
- Rather, gyms with oligopolistic market structures would adopt non-price strategies to differentiate their products from other gyms.
 - This could include advertising to increase demand for their gym services and make demand more price inelastic, thereby raising revenue and hence profits (assuming advertising costs do not exceed the rise in revenue), or by developing new “products” e.g. exercise classes or offering high-tech equipment to achieve the same effect.
 - *Note: Alternative strategies for oligopolies include collusion, process innovation etc.*
- **[Situation]** That said, not all gyms fall cleanly under the oligopolistic market structure. Smaller boutique gyms like Ritual could be argued to be monopolistically competitive. These boutique gyms face relatively lower BTEs, as their floor space is usually smaller and hence rental costs lower, and they tend to not require as much gym machinery.
 - That said, monopolistically competitive firms typically have some degree of local monopoly power, where they have some ability to set price within a localised market or a given niche. For example, smaller boutique gyms may bank on consumers' desire for unique service offerings such as specialised fitness programmes or niche classes (e.g. sound bath yoga, animal flow classes), to attract specific customer segments.
 - While also a form of product differentiation, such non-price strategies are likely to be different from that adopted by larger chain gyms in oligopolistic market structures. Furthermore, having some degree of local monopoly power, monopolistically competitive gyms may be able to engage in price strategies such as price discounts.
 - *Other aspects of product differentiation include: customer service experience, brand-building to foster brand loyalty.*
- As such, depending on the market structure of the gym, they may opt to adopt different strategies to remain in the market.

Key Argument 2: Explain another factor that a gym would consider when deciding which strategies to adopt to remain in the market.

- **Another factor that a gym would consider when deciding which strategies to adopt to remain in the market would be the ability of the gym to secure funding.**
- If the gym were earning supernormal profits over a period of time (i.e. accumulated financial reserves) or is able to secure funding from outside sources (e.g. investors or bank loans), this would increase the range of strategies available to it, as it would be able to utilise strategies that require significant financial outlay.
 - These include strategies such as product and process innovation, as well as advertising, that require significant initial spending by the firm to develop new and/or better-quality products, to develop more efficient production processes, as well as to promote their products.
- In the case of product innovation and advertising, assuming the strategy is successful, there would be an increase in the demand.

- On the other hand, if the gym had been earning subnormal or normal profits over a period of time or is lacking in funding ability, they would be less able to adopt strategies that require significant costs (such as those explained above), as they would be less likely to have accumulated financial reserves.
- Instead, such gyms may have to resort to strategies that do not require as much financial input – mostly price strategies. An example of such a strategy would be that of third-degree price discrimination, where the gym charges different prices to different groups of consumers for the same gym services, for reasons not associated with differences in the cost of production.
 - By charging lower prices to the group with more price elastic demand (e.g. students) and higher prices to the group with less price elastic demand (e.g. adults), gyms can earn higher revenue from both groups and hence higher profits, thus increasing their ability to remain in the market, assuming costs remain constant.
- Hence, the level of profits earned by the firm affects the types of strategies it can adopt to remain in the market.

Alternative factor:

- **Another factor that a gym would consider when deciding which strategies to adopt to remain in the market would be that of the primary challenge faced (and its root cause).**
- The types of strategies that a gym would choose to adopt would have to take into account the economic and market conditions at the time, as well as their root causes.
- From (a), the gym industry was facing several challenges that led to both a fall in demand for its services as well as a rise in the cost of production. A gym might choose to respond to a fall in demand due to COVID-19 or changing workout fads by pivoting to strategies that might better cater to the tastes and preferences of consumers.
 - For example, due to the COVID-19 pandemic, consumers might increasingly prefer to exercise in the comfort of their own homes instead. As such, a gym could offer workout regimes on an online platform instead to raise the demand for its services (as many gyms did during the pandemic).
- On the other hand, if the primary external challenge facing the gym was a rise in costs instead, perhaps the most suitable strategy for the gym to implement would be process innovation to find a way to produce most efficiently instead (e.g. replacing some labour with technology).
 - This could help to lower its production costs, thereby raising profits and helping it to remain in the market.
- As such, a gym would need to consider the most major challenge it faces when deciding on what strategy to employ to remain in the market.
- *Other acceptable factors: Types / extent of government regulation (e.g. licenses, price regulation)*

Evaluative Conclusion (a reasoned stand that follows from the above analysis + 2 well-developed ATMS evaluative angles):

- Overall, a gym would need to consider several factors in deciding which strategies to use to remain in the market, with all of them being important in the decision. **[Stand]** Perhaps the foremost factor it needs to consider is its funding ability, followed by its market structure.
- *Note: Any other chosen “stance” is acceptable, as long as it can be justified below.*
- **[Magnitude/Situation]** The gym should first and foremost consider its financial ability, either through accumulated supernormal profits or through securing funding, as this would affect the financial capability of the firm to implement expensive strategies such as product and process innovation.
 - The lack of funding ability or accumulated financial reserves would restrict the strategies available to the firm in the first place. With this in mind, the gym can then go ahead to consider its market structure in the gym industry, particularly in terms of the extent of interdependence with other gyms like it.
 - Smaller boutique gyms would likely benefit from some sort of monopoly power and hence can focus on ensuring differentiation to reduce the PED of their services, while larger chain gyms need to greater take into account the actions of their rivals.
- **[Situation]** That said, given that the gym’s short-term goal is to remain in the market, it should take into consideration the challenges it is facing, identify its primary challenge(s) and establish the root cause(s), such that it can respond by employing strategies that target the root cause(s) of the challenges.

Mark Scheme

Level	Knowledge, Application/Understanding and Analysis	Marks
L3	For a well developed answer that has: • Good scope – explains at least two factors that affect a gym’s decision on which strategies to adopt to remain in the market; and • Good rigour – explains the links to how the factors affect a gym’s choice(s) of strategies; and • Good application to the context of gym market	8 – 10
L2	For an answer that: • Lacks scope – only explains one factor that affect a gym’s decision on which strategies to adopt to remain in the market; and/or • Lacks rigour – there are gaps in explaining the links to how the factors affect a gym’s choice(s) of strategies; and/or • Lacks application to the context of to the context of gym market	5 – 7
L1	For an undeveloped answer that: • Is descriptive, lacking in application of economic theory; and/or • Contains serious and pervasive conceptual errors; and/or • Is largely irrelevant.	1 – 4
Evaluation		
E3	For an answer that provides supported evaluative statements on the relative importance of the factors explained and arrives at an analytically well-reasoned and synthesised judgement with reference to the context (i.e. gym industry).	5
E2	For an answer that makes some attempt at evaluating the relative importance of the factors explained, but evaluative comments may be limited in scope, or not fully elaborated upon.	3 – 4
E1	For an answer that gives superficial evaluative statement(s) without supporting analysis and elaboration.	1 – 2

3 Excessive tourism often leads to significant water and air pollution, as increased demand for resources and services strains local environments, leading to inefficiencies and long-term ecological damage.

- (a) Explain how a rational consumer decides whether to travel abroad for leisure, and how such decision-making may lead to an inefficient allocation of resources. [10]
- (b) Discuss the measures a government can implement to improve resource allocation in the market for tourism services. [15]

Part (a) Question interpretation

Command word/phrase	Explain how	Explain the process using economic theory and relevant examples where appropriate
Content	<i>Rational consumer decide to travel abroad for leisure</i> <i>Such decision-making can lead to allocative inefficiency</i>	Maximise self-interest of utility/satisfaction. Weigh MPC and MPB Based on preamble, excessive tourism brings about negative spillover effects to third parties.
Context	<i>Travel abroad for leisure</i>	To consider the context of tourism

Students are to explain (1) rational decision-making by consumers – consumers are willing to buy an additional unit of travel so long as the additional benefit they derived from travelling (marginal private benefit, MPB) exceeds the additional cost (MPC) they incur from buying: the price of travelling and (2) how rational decision making by consumers will result in allocative inefficiency because of the externalities associated with tourism.

Introduction

- Consumers make rational decisions where they aim to maximise their self-interest.
 - Marginal benefit (MB) is the additional benefit derived from doing one more unit of an activity.
 - Marginal cost (MC) is the additional cost derived from performing one more unit of an activity.
- To maximise their self-interest, rational consumers apply the marginalist principle, which states that the optimal level of an activity is attained when $MB = MC$, and consumers maximise their net total private benefits.
- In the pursuit of self-interest, consumers consider only their private costs and private benefits.

Body

Key Argument 1: Explain rational decision-making by consumers – consumers are willing to buy an additional unit of travel so long as the additional benefit they derived from travelling (marginal private benefit, MPB) exceeds the additional cost (MPC) they incur from buying: the price of travelling. When this occurs, the consumer maximises his net total private benefit.

- **Consumers seek to maximise net total private benefit (consumer surplus) from consuming goods and services.**
- Consumers derive utility or satisfaction from traveling to a foreign destination. They are willing to take an additional trip abroad as long as the additional benefit they gain from the experience (marginal private benefit, MPB) – in this case, the added enjoyment from sightseeing, relaxation or experiencing new cultures – exceeds the additional cost they incur, which is the price of the trip (e.g. flight tickets, accommodation, tourist attractions).
 - When this occurs, the consumers maximise their net total private benefit from this leisure travel.

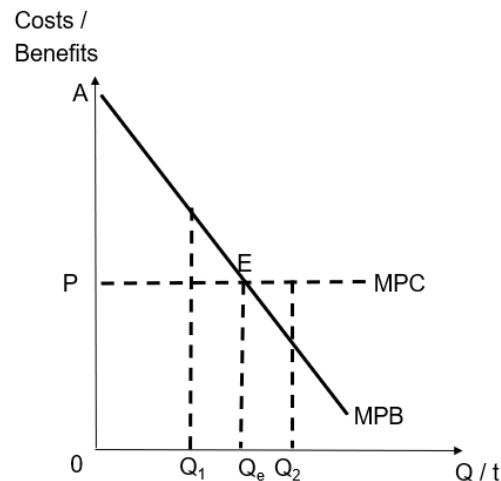


Figure 1: Self-interest maximisation by a consumer

- According to the marginalist principle, as long as the marginal private benefit (MPB) of traveling abroad is greater than the marginal private cost (MPC), the consumer derives a net private benefit from doing so and should continue to engage in more leisure travel, as this would increase his net total private benefit.
- With reference to Figure 1 above, the MPB curve is downward-sloping due to the law of diminishing marginal utility. This law states that after a certain level of travel experiences, the additional benefit derived from each subsequent trip declines as more trips are taken.
- If the price of traveling abroad is P, the consumer will travel up to Q_e trips, the point where his MPB equals MPC.
 - For any number of trips below Q_e , for example at Q_1 , MPB is greater than MPC, the rational consumer will choose to travel because the additional private benefit exceeds the additional private cost, indicating that net total private benefit is rising.
 - On the other hand, a rational consumer will not travel if MPB is less than MPC, as in the case at Q_2 , since the additional private benefit would be lower than the additional private cost, causing net total private benefit to fall.
- Consumers will aim to travel up to Q_e trips, where MPB intersects MPC, because at this point, the consumer maximises net total private benefits (consumer surplus).
 - The area representing net total private benefits, AEP, is calculated by the difference between the total private benefit (area $0AEQ_e$) and total private cost (area $0PEQ_e$).
- Therefore, when the consumer travels up to Q_e trips, he will maximise his net private benefit.
- The consumer must also consider his income constraint before deciding whether to travel abroad. If his income is insufficient, he would be unable to afford the trip, even if the MPB exceeds the MPC.

Kay Argument 2: Explain how rational decision making by consumers will result in allocative inefficiency because of the externalities associated with tourism.

- **Market failure (and hence allocative inefficiency) occurs in the market for tourism services due to the presence of negative externalities arising from this consumption activity.**
- Negative externalities arise when the actions of consumers impose spillover costs on third parties who are not directly involved in the economic transaction.
- **[ET]** When more people travel abroad for leisure, this can lead to overcrowding in popular tourist destinations, contributing to the degradation of local environments and infrastructure.
 - Local residents, who are third parties not involved in the consumption of leisure travel in their home country, may experience a decline in their quality of life due to increased pollution, noise, and strain on public services such as waste management or public transport.
 - Moreover, the environmental costs of tourism can extend to global issues like increased carbon emissions from air travel. These emissions contribute to climate change, affecting people worldwide, even though they are not directly involved in the tourists' decision to travel.
- As consumers are only concerned with maximising their net private benefit when traveling abroad and consume tourism services, they disregard these external costs imposed on others.

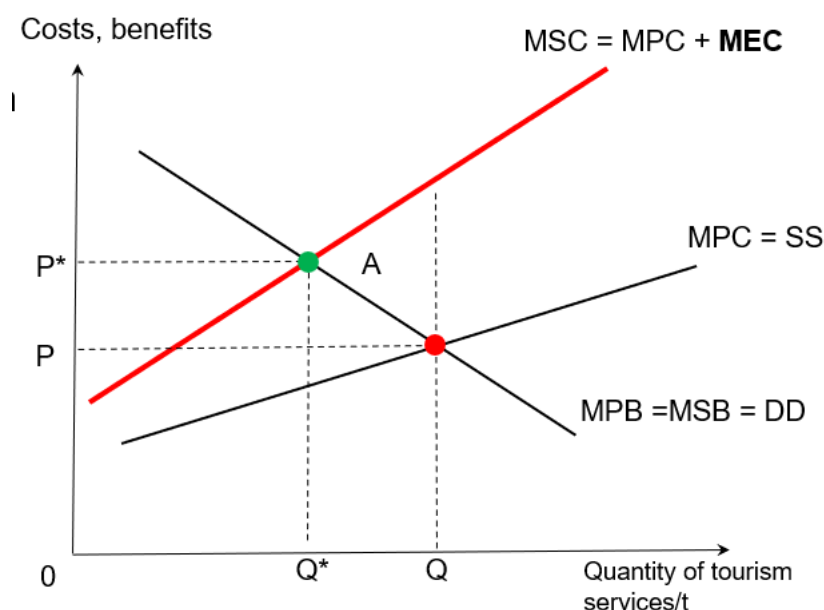


Figure 2: Allocative inefficiency in the Market for Tourism Services

- *Note: In Figure 2, the MSC and MPC curves can be illustrated as parallel to one another instead.*
- **[D]** The negative externality causes the marginal social cost (MSC) to be greater than the marginal private cost (MPC) from consuming tourism services since the additional cost to society (MSC) include both the additional private cost to consumers (MPC) as well as the additional costs to third parties not involved in the consumption (MEC).
 - MSC curve is higher than the MPC curve, where MEC is the vertical distance between the MSC and MPC curves.
- **[QQ]** Under the free market, consumers will base their decisions on their MPC and disregard the external costs. Assuming there are no positive externalities, $MPB = MSB$.

- When left to market forces, the market equilibrium output will be at Q where demand equals supply (which is also where $MPB = MPC$).
- However, the socially optimal output is at Q^* , where $MSB = MSC$, which is where society's welfare is maximised.
- Hence, under the free market, there is overconsumption of tourism services by $(Q - Q^*)$ units, resulting in welfare loss to society.
- **[D]** For every unit between Q^* and Q , total social costs (area under MSC) is greater than total social benefits (area under MSB); consuming one more unit of tourism services thus adds more to total social cost than to total social benefit.
 - The overconsumption of QQ^* units thus lead to a welfare loss of area A .
- *Note: Answers that focus on overconsumption of air travel resulting in negative externalities associated with air travel e.g. environmental pollution will also be accepted. Do note though that the question is on travel/tourism **in general**.*

Conclusion

- In conclusion, rational consumers' decisions on whether or not to travel abroad for leisure may lead to an inefficient allocation of resources due to consumers' disregard for the negative externalities generated.

Mark Scheme

Level	Knowledge, Application/Understanding and Analysis	Marks
L3	For a well developed answer that has: • Good scope – explains the application of marginalist principle when a rational consumer decides whether to travel, and how such decision-making may lead to an inefficient allocation of resources; and • Good rigour – supporting benefit and costs analysis with accurate and well-labelled diagram(s); and • Good application to the context of tourism market	8 – 10
L2	For an answer that: • Lacks scope – only explains the application of marginalist principle when a rational consumer decides whether to travel, or how such decision-making may lead to an inefficient allocation of resources; and/or. • Lacks rigour – there are gaps in using benefits and costs analysis and/or explanation is not supported by accurate and well-labelled diagrams(s) where relevant; and/or • Lacks application to the context of market for tourism	5 – 7
L1	For an undeveloped answer that: • Is descriptive, lacking in application of economic theory; and/or • Contains serious and pervasive conceptual errors; and/or • Is largely irrelevant.	1 – 4

- 3 Excessive tourism often leads to significant water and air pollution, as increased demand for resources and services strains local environments, leading to inefficiencies and long-term ecological damage.

(b) Discuss the measures a government can implement to improve resource allocation in the market for tourism services. [15]

Part b Question Interpretation

Command word/phrase	<i>Discuss the measures</i>	To examine and present the different aspects of the measures and give a judgement on these measures
Content	<i>Measures to improve resource allocation</i>	Policies that government can implement to address allocative efficiency. Need to explain the workings of proposed policies and their limitations/unintended consequences
Context	<i>Market for tourism services</i>	To apply to context of tourism market

Students are to explain two measures and their limitations/unintended consequences that a government can implement to improve resource allocation in the market for tourism services and come to a judgement on these measures.

Introduction

- As explained in (a), tourism services often generate significant negative externalities, such as environmental degradation, overcrowding, and strain on local infrastructure, leading to overconsumption and market failure.
- In response, governments may implement corrective policies such as a tourism tax or quotas to force consumers to internalise these external costs and restore allocative efficiency.
- This essay will assess the appropriateness of such policies in addressing negative externalities in the tourism sector, focusing on the sources of external costs and how well these measures can correct the market failure while considering potential unintended consequences on the industry and local economies.

Body

Key Argument 1: Explain one measure that a government can implement to improve resource allocation in the market for tourism services and its limitation/unintended consequence

- A tax on tourism services is a tax levied on the producers of tourism-related activities or services to manage over-tourism.**
 - For example, a fee is imposed when tourists visit popular tourist attractions, such as national parks, historical monuments, or cultural heritage sites.
- The tax imposed will reduce the overconsumption of tourism services as tourists internalise the external costs associated with tourism.
- This tax is designed to be equal to the marginal external cost at the socially optimal level, Q^* .
 - The tax will cause the marginal cost of production of hotel operators or operators of tourist attractions to rise.

- By raising the MPC to $MPC' = MPC + \text{tax}$, the tax forces tourists to internalise the external costs, such as environmental degradation, overcrowding, and strain on local infrastructure, which are typically borne by third parties.
- As the rise in MPC leads to higher prices of tourism services from P to P^* , the quantity demanded for tourism services by consumers will fall.
- As a result, the new private equilibrium level of tourism where $MPC' = MPB$, Q' now coincides with the social equilibrium at Q^* , helping to reduce the overconsumption of tourism services and resolve the market failure of allocative inefficiency.

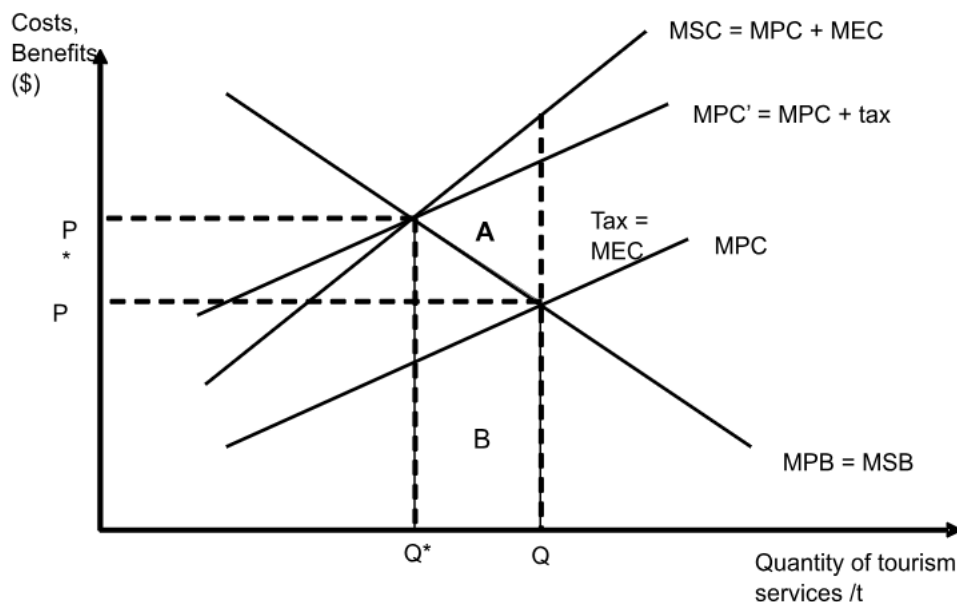


Figure 3: Production Tax

- Note: Similarly, in Figure 3, the MSC and MPC curves can be illustrated as parallel to one another as well.

Limitation/unintended consequence:

- However, governments often face imperfect information regarding the true extent of the external costs associated with tourism, making it challenging to determine the correct level of tax to impose.
- If the tax is set too low, it may not sufficiently reflect the external costs, leading to continued overconsumption.
 - For example, tourists may continue visiting excessively, contributing to environmental damage or strain on local resources, thereby not correcting the market failure.
- On the other hand, if the tax is set too high, it may overly deter tourists from visiting. This could result in an underconsumption of tourism services, leading to a different type of welfare loss to society.
 - Tourism-dependent businesses, employees, and local economies could suffer from reduced demand, causing negative effects on employment and income.
- If this new welfare loss from underconsumption exceeds the size of the original welfare loss from overconsumption, it could lead to government failure.

OR

- In addition to this, the **price elasticity of demand (PED)** for tourism services plays a crucial role in determining the effectiveness of the tax in reducing overconsumption.
 - If the demand for tourism services is relatively **price inelastic**, meaning that consumers are less sensitive to changes in price, then a tax may not significantly reduce the quantity of tourism demanded.

- Wealthier tourists, for instance, may continue to consume tourism services despite higher costs, meaning the negative externalities, such as environmental degradation and overcrowding, may persist.
- This undermines the policy's effectiveness in correcting the market failure.

Key Argument 2: Explain another measure that a government can implement to improve resource allocation in the market for tourism services and its limitation/unintended consequences

- **A tourism quota is a regulatory measure implemented by governments to limit the number of tourists allowed to visit specific destinations or participate in certain tourism activities within a set period of time.**
- In Thailand, the government implemented a tourism quota at Maya Bay to protect the environment after severe degradation caused by overcrowding.
 - The bay was initially closed for several months to allow for recovery, and when reopened, a strict limit of 375 tourists per hour was enforced to prevent further damage to coral reefs and marine life.
- When a tourist quota is imposed to reduce allocative inefficiency in the market for tourism, this measure effectively places a cap on the quantity of tourism services that can be consumed.
- The quota limits the quantity of tourism services to the socially optimal level, where marginal social costs (MSC) is equal to the marginal social benefits (MSB) at Q^* .

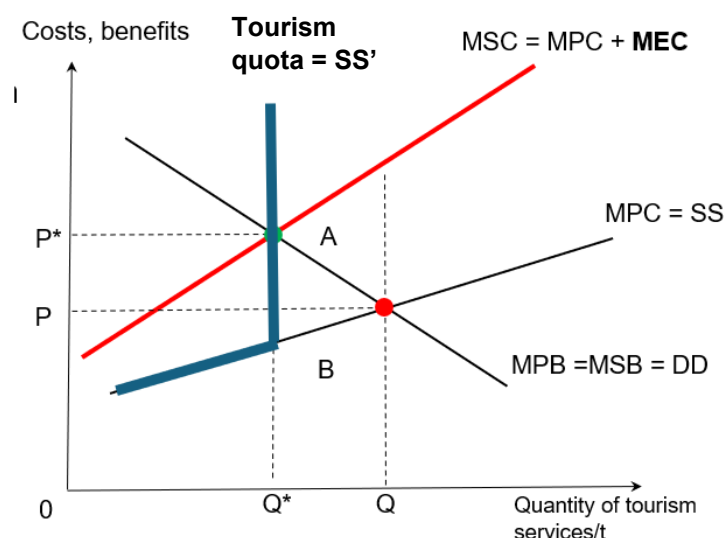


Figure 4: Tourism Quota

- This quota is represented as a vertical line at Q^* indicating that tourism consumption cannot exceed this level.
 - The quota effectively caps the quantity of tourism services supplied, regardless of demand. The supply curve becomes perfectly inelastic at the quota level Q^* .
- The 'new supply curve' is represented by the bold line SS' and equilibrium price of tourism services is at where $SS' = DD$ at P^* and this quantity occurs at the socially optimal output level, thus eliminating the welfare loss to society and restoring allocative efficiency in the market for tourism services.

Limitation/unintended consequence:

- Tourism quotas may not effectively address the root causes of the environmental issues associated with tourism.

- Quotas only limit the overall number of tourists rather than targeting the specific activities that generate the most significant negative externalities.
 - For example, in a coastal destination, restricting the number of visitors may do little to mitigate the environmental damage caused by large cruise ships that discharge waste into the ocean or by high-pollution transportation methods such as tour buses that emit substantial carbon emissions.
 - These high-impact activities can continue to degrade the environment even with fewer tourists, meaning the quota fails to correct the true source of market failure.

OR

- Implementing and enforcing tourist quotas can be an ineffective solution for addressing negative externalities in the tourism market due to high administrative costs involved.
- Developing an effective system for issuing permits and regulating tourist numbers involves enforcement officers and technological infrastructure, and this system may be costly for the government.
- Governments must also monitor visitor flows, often across multiple access points, which can make tracking and enforcement particularly challenging, especially for larger countries. This increases the risk of tourists bypassing the quota system, either intentionally or due to weak enforcement, which would undermine the policy's effectiveness.

OR

- Tourist quotas may worsen income inequality and impede economic growth in regions that are heavily dependent on tourism as a primary source of income, such as Bali in Indonesia.
- By limiting the number of tourists permitted in high-demand destinations, the policy can reduce income streams for small businesses, tour operators, and local communities and widens income inequality between different consumer groups in the country.
- The reduced tourist expenditure may also cause a fall in exports of tourism services and may stifle economic growth when AD falls, *ceteris paribus*. These regions may also struggle to attract alternative forms of investment and employment, thus hampering sustained economic growth.
- In essence, tourist quotas can deepen income inequality and hinder economic development in vulnerable tourism-dependent regions.

Alternative Policy

- Public awareness campaigns aim to educate tourists about the environmental and social impacts of their travel behaviours.
- These campaigns can encourage responsible tourism practices such as reducing waste and avoiding over-touristed areas.
 - Governments can disseminate information through various channels, including social media or travel agencies. By increasing awareness, the campaigns can shift consumer preferences towards more sustainable tourism choices, such as eco-friendly accommodations or low-impact travel methods.
 - For example, tourists might opt to visit less crowded destinations or choose tour operators that prioritise sustainability.
- When more tourists choose to practice eco-tourism or travel sustainably, MPC will move closer to MSC, as now the extent of external cost to third parties is smaller (MEC becomes smaller).
- *Note to students: The public awareness campaigns explained above have the intended effect of **moral suasion**, i.e. to encourage/persuade consumers to take into account the negative externalities in their consumption of tourism services generates on the environment. This should be articulated clearly, as the source of market failure here arises from consumers **disregarding** the external costs they impose on others, not from consumers being unaware of these external costs.*

Evaluative Conclusion *(a reasoned stand that follows from the above analysis + 1 well-developed ATMS evaluative angle would suffice):*

- Whether taxes on tourism services and quotas are effective in addressing the market failure problem will depend on the nature of tourism services and the nature of the country.
- **[Stand + Situation/Magnitude]** Taxes on tourism services are generally more effective for broad-based, high-volume tourism services such as air travel, hotel stays, or access to popular tourist cities.
 - Large numbers of tourists often consume these services, making it easier to apply a tax uniformly. When the externality is relatively diffused, such as carbon emissions from air travel or general congestion in a city, a tax on tourism services can be an efficient way to internalise these costs across all users.
 - The tax distributes the burden across many consumers and raises revenue that can be reinvested in mitigating the externalities, such as environmental protection or infrastructure upgrades.
- **[Stand]** Quotas are more effective in limiting access to specific destinations or tourist attractions where the externalities are severe and concentrated.
 - For example, world heritage sites, national parks, or protected ecosystems such as coral reefs may experience irreversible damage if visitor numbers exceed sustainable limits. In such cases, a tourist quota ensures that the number of tourists remains within the site's ecological carrying capacity.
- **[Stand + Situation]** Taxes on tourism services are often more effective in developed countries with established tax systems, strong administrative capabilities, and diverse tourism markets. These countries tend to have the infrastructure to efficiently collect taxes and reinvest the revenue into public services, infrastructure, or environmental conservation.
 - In countries where tourism is a large part of the economy and involves many different sectors (e.g., hotels, airlines, restaurants), a tax can help manage the negative externalities across the whole economy without restricting tourism excessively.
- **[Stand]** Countries with a few key attractions that are particularly sensitive to over-tourism (e.g. small island nations or countries with unique ecosystems) may benefit more from quotas. These countries may prefer to cap the number of visitors to protect their natural and cultural resources rather than imposing taxes, which may not effectively reduce tourist numbers in ecologically sensitive areas.

Mark Scheme

Level	Knowledge, Application/Understanding and Analysis	Marks
L3	Well-explained and analytical answer on at least two measures that are supported with diagrammatic illustrations. For a well developed answer that has: • Good scope – explains at least two measures that can used to achieve allocative efficiency and their limitations/unintended consequences; and • Good rigour – supporting benefits and costs analysis with accurate and well-labelled diagram(s); and • Good application to the context of tourism market	8 – 10
L2	For an answer that: • Lacks scope – only explains how one measure that can used to achieve allocative efficiency and their limitations/unintended consequences; and/or • Lacks rigour – there are gaps in using benefits and costs analysis and/or explanation is not supported by accurate and well-labelled diagrams(s) where relevant; and/or Lacks application to the context of tourism market	5 – 7
L1	For an undeveloped answer that: • Is descriptive, lacking in application of economic theory; and/or • Contains serious and pervasive conceptual errors; and/or • Is largely irrelevant.	1 – 4
Evaluation/Synthesis		
E3	For an answer that provides supported evaluative statements on the relative effectiveness and/or appropriateness of the measures explained, and arrives at an analytically well-reasoned and synthesised judgement with reference to the context (i.e. market for tourism services).	5
E2	For an answer that makes some attempt at evaluating effectiveness and/or appropriateness of the measures explained, but evaluative comments may be limited in scope, or not fully elaborated upon.	3 – 4
E1	For an answer that gives superficial evaluative statement(s) without supporting analysis and elaboration.	1 – 2

4 Governments seek to attain various macroeconomic goals, one of which is inclusive and sustainable economic growth, in order to raise the living standards of its citizens.

- (a) **Explain why governments aim to achieve inclusive and sustainable economic growth.** [10]
- (b) **Discuss the extent to which policies to achieve inclusive economic growth would cause difficulties for an economy.** [15]

Part (a) Question Interpretation

Command word/phrase	<i>Explain why</i>	To utilise economic reasons and reasoning to explain why governments aim to achieve inclusive and sustainable economic growth.
Content	<i>Aim</i> <i>Inclusive and sustainable economic growth</i>	Goals of the government → this is a consequence question which requires you to explain the benefits / positive consequences Inclusive growth → benefits of growth are spread evenly with little income inequality Sustainable growth → minimal impact on environment for future generations
Context	<i>Governments</i>	Different countries' governments

This answer requires an explanation of the benefits of achieving inclusive and sustainable economic growth, thus justifying why governments aim to achieve these two types of growth.

Introduction

- Governments generally aim to achieve various macroeconomic goals, of which inclusive and sustainable economic growth is just two of them.
- Inclusive growth indicates a rate of growth that is sustained over a period of time, broad-based across economic sectors, and creates productive employment opportunities for the majority of the country's population. It takes income distribution into consideration and does not contribute to worsening income inequality.
- Sustainable growth indicates that there must be a sustained rate of growth without creating other significant economic problems (such as depleted resources and environmental problems), particularly for future generations.

Key Argument 1: Governments aim to achieve inclusive economic growth because of the benefits it brings to society.

- **Achieving inclusive economic growth brings about higher material and non-material living standards for the citizens.**
- Inclusive growth requires economic growth to be sustained first, which refers to long-run, non-inflationary growth. When there is sustained growth, real GDP / income rises and assuming population size is constant, real GDP per capita rises, thus household's income and purchasing power increases, increasing the ability of consumers to purchase goods and services. Citizens' material living standards rises as a result.
- However, economic growth may not always be inclusive. Growth driven by technological advancement could lead to the decline of some industries and redundancy of some jobs.
 - For instance, many supermarkets are utilising self-check-out counters which displaces cashiers. If these unemployed workers lack the relevant skills to enter expanding sectors,

they will become structurally unemployed. Such growth will worsen income inequality, possibly resulting in social unrest, and lowering non-material living standards. Inclusive growth, in ensuring the benefits of economic growth are distributed among most of the population, will improve income equality and social stability, thus increasing citizens' non-material living standards.

- Additionally, boosting the income of low-income individuals can further promote economic growth. This is because the marginal propensity to consume domestic goods for low-income households is high.
 - When the incomes of low-income households rise, there will be a rise in Consumption expenditure, as these households will likely spend most if not all their increased income on domestic goods and services, thus increasing AD and achieving actual growth and further increases in material living standards.
 - At the same time, as there are productive employment opportunities for majority of the population, unemployment rates and crime rates fall due to higher opportunity cost of crime.
 - This could lead to an improvement in the non-material aspect of SOL as residents feel more secure both in employment and in their society.

Key Argument 2: Governments aim to achieve sustainable economic growth because of the benefits it brings to society.

- **Achieving sustainable economic growth brings about higher material and non-material living standards for the citizens.**
- As explained earlier, sustained economic growth in the first place can lead to a continued increase in material living standards of a country over time. By attaining sustainable growth too, it means that natural resources are not depleted and not creating environmental problems for future generations, and this will improve the quality of environment, thus non-material standard of living improves.
- Since not all natural resources are renewable, exploitation of these resources to bring about economic growth could mean insufficient natural resources for future generations to produce goods and services.
 - When the country's natural resources become depleted, the country's productive capacity shrinks where the vertical segment of AS shifts left over time. This leads to a fall in real GDP and hence lowers material standards of living of the future generation.
 - Achieving sustainable economic growth is therefore able to ensure future economic growth as it will ensure sufficient resources for the country's productive capacity to rise in the future. This will ensure higher material living standards in future.
- In addition, destruction of the natural environment tends to occur alongside economic growth. Profit-maximising firms typically disregard external costs that arise from pollution created during the production process in their output decisions.
 - The more rapid the economic growth, the more waste (i.e. air and water pollution from production) the environment must absorb. Rapid economic growth may thus lead to a depletion of clean water and breathable air for residents in the country. Future generations will then suffer a decline in their living standards.
 - Therefore, with sustainable economic growth, there will be better sanitation, clean water for and breathable air for future generations, thus ensuring higher non-material living standards.

Mark scheme

Level	Knowledge, Application/Understanding and Analysis	Marks
L3	For a well-developed answer that has: ● good scope – explains benefits of both inclusive and sustainable economic growth; and ● good rigour – with good linkages to material and non-material living standards; and ● good use of examples – to explain the benefits of the two types of economic growth.	8 – 10
L2	For an answer that: ● lacks scope – only explains either the benefits of inclusive or sustainable economic growth; ● lacks rigour – there are gaps in the analysis, or did not link the benefits to material and non-material living standards. Answer lacks examples to link to the two types of economic growth.	5 – 7
L1	For an undeveloped answer that: ● is descriptive, lacking in application of economic theory; and/or ● contains serious and pervasive conceptual errors; and/or ● is largely irrelevant.	1 – 4

4 Governments seek to attain various macroeconomic goals, one of which is inclusive and sustainable economic growth, in order to raise the living standards of its citizens.

- (b) Discuss the extent to which policies to achieve inclusive economic growth would cause difficulties for an economy. [15]**

Part (b) Question Interpretation

Command word/phrase	<i>Discuss the extent</i>	To examine and present the different aspects of a problem or subject, and give a judgement on the extent to which policies for inclusive growth could cause difficulties for economy
Content	<i>Policies to achieve inclusive growth...</i>	Transfer payments / redistributive policies and supply-side policies
	<i>Cause difficulties for economy</i>	Potential unintended consequences / trade-offs for the economy (<i>as the balance to the policies</i>)
Context	<i>An economy</i>	No specified economy or country. Students to come up with examples of their own

A relevant response requires students to first explain how two policies would work to achieve inclusive growth, before explaining the potential negative unintended consequences (difficulties) on the economy as its balance. The evaluation entails a weighing of which policy will or will not cause eventual difficulties for an economy.

Introduction

- The government's macroeconomic goals include sustainable and inclusive economic growth, low unemployment, price stability and healthy balance of trade. However, due to certain conditions, the pursuit of these goals may lead to unintended consequences of the economy.
- This essay aims to discuss two policies to achieve inclusive economic growth and their accompanying unintended consequences on the economy.

Body

- **Policy 1: Governments can increase its transfer payments to lower income households and raise income taxes on higher-income households to achieve inclusive growth.**
 - One possible policy is via an increase in income taxes on higher income households and / or increase transfer payments to lower income households to reduce the income gap, thus achieving inclusive growth.
 - In Singapore, there is a progressive income tax system where higher income individuals are taxed a higher proportion of their income while lower income individuals are taxed a lower proportion of their income. This reduces the income disparity between the high-income households and low-income households, thus improving income equality.
 - Furthermore, the tax revenue collected can then be used to provide transfer payments to the lower income households in various forms such as means-tested subsidies for education, housing and healthcare.
 - This will also ensure that the lower income households will have sufficient amount to ensure that their basic needs are met, therefore ensuring inclusive economic growth.
- **However, such a policy may cause difficulties for the economy.**
 - An increase in transfer payments require expenditure by the government and needs to be financed. If the government did not increase the income tax on higher income individuals, then it will suffer a worsening budget position, which may not be sustainable in the long run.
 - As such, the government must collect more tax revenue in order to achieve inclusive growth while maintaining fiscal sustainability.
 - Raising income taxes has contractionary effects on the economy, as the increase in personal income tax will reduce Consumption expenditure and thus decreasing AD and real incomes.

**Alternatively, students can also explain the unintended consequence of demand-pull inflation should the increase in C via transfer payments rise while economy is near full employment.*

- **Policy 2: Governments can also utilise supply-side policies like training and retraining of the lower skilled and lower income workers to achieve inclusive growth.**
- Another possible policy is to increase spending on education and upskilling of the lower-skilled workers to achieve inclusive growth.
- The subsidisation of education and skills upgrading allows for an increase in the quality of labour.
 - This increases the productive capacity directly, thus increasing LRAS. The higher the labour productivity, the unit cost of production lowers too, thus increasing SRAS.
 - The rightward and downward shift of the AS curve leads to lower general price levels and higher real GDP, achieving sustained growth in the country.
- Furthermore, this economic growth becomes inclusive, should the upskilling be targeted at the lower skilled and lower income workers in the economy. In Singapore, there is the SkillsFuture Credit which provides a subsidy for workers who go for training and upskilling and encourages them to learn new skills to increase their productivity.
 - By upgrading the skills of lower skilled workers, they become more productive and will be more attractive to firms, which will be more willing to pay higher wages to these better skilled workers since these workers can now help the firms earn more revenue.
 - This will contribute to the reduction of the income gap and also increases occupational mobility as people are more able to pick up new jobs when the old ones are lost or switch to the growing job sector.
 - For those workers who are unemployed, helping them pick up a new skill for the next job would enable them to earn an income and narrows the income disparity, achieving inclusive growth.

- **However, such a policy may cause difficulties for the economy.**
- Supply-side policies like these also require usage of government's budget. This means that the government may have lesser funds to allocate to carry out expansionary fiscal policy to boost further growth for example.
 - Furthermore, should the policy be successful to increase productive capacity, there may be a greater extent of spare capacity. If there is no accompanied increase in real output, this may increase unemployment in the economy.

Evaluative Conclusion (a reasoned stand that follows from the above analysis + 1 well-developed ATMS evaluative angle would suffice):

- The extent to which economic policies used to achieve inclusive growth will cause difficulties for the economy depends on the type of policies implemented, the state of economy as well as the ability of the government to implement mitigating policies to address these difficulties.
- **[Stand / Situation + Magnitude]** The potential benefits of supply-side policies will mean that there may be lesser difficulties for the economy. This is because, supply-side policies such as skills retraining could generate sustained growth as well, thus resulting in lesser difficulties for the economy. In contrast, using just transfer payments can rake up government debt and disincentivise investments and there may not be any additional benefits beyond merely achieving inclusive growth.
- **[Situation]** If the economy is already operating close to productive capacity, achieving inclusive growth via transfer payments will likely cause demand-pull inflation. Governments that are in heavy fiscal debt (e.g. Sri Lanka) may also find themselves facing the unintended consequence of negative growth as a result of reduced confidence in the economy, thus limiting the effectiveness of the policies implemented in the first place.
- **[Situation]** If the country is currently experiencing high levels of structural unemployment, having targeted skills training will ensure lesser difficulties because it directly helps the lower skilled workers find employment, and close the income gap. Merely giving transfer payments to the lower income does not tackle the root cause of the problem and may lead to further difficulties.
- **[Alternative]** By choosing appropriate policies that suit the country's context, it is possible to ensure that the achievement of inclusive growth does not incur trade-offs with other macroeconomic goals in the longer term.
 - Therefore, if the governments are aware of the unintended consequences of pursuing inclusive growth on other macroeconomic goals, they can implement appropriate complementary policies to ensure that the growth is inclusive while alleviating other macroeconomic problems that might result e.g. by implementing supply-side policies that increase the productive capacity of the economy and hence attain potential growth to alleviate the demand-pull inflation that results from spending on transfer payments.
 - Fiscally, it is perhaps more sustainable to rely on supply-side policies such as subsidising skills upgrading and training programmes in addition to transfer payments to allow lower income households earn higher wages over time. They can then stop receiving the transfer payments while also contributing to the tax revenue collected.

Mark Scheme

Level	Knowledge, Application/Understanding and Analysis	Marks
L3	For a well-developed answer that has: • good scope and balance – explains at least 2 policies to achieve inclusive growth; and their accompanying unintended consequences; • good rigour – clearly explains how policies work, together with their unintended consequences using appropriate macroeconomic analysis; • good application to context – supports analysis with real world examples of policies used to achieve inclusive growth.	8 – 10
L2	For an answer that: • lacks scope and balance – only explains one policy to achieve inclusive growth together with their unintended consequences, or only the workings of two policies without their unintended consequences; • lacks rigour – incomplete use of macroeconomic analysis; • lacks application – does not apply analysis to the context of inclusive growth.	5 – 7
L1	For an undeveloped answer that: • is descriptive, lacking in application of economic theory; and/or • contains serious and pervasive conceptual errors; and/or • is largely irrelevant.	1 – 4
Level	Evaluation/Synthesis	Marks
E3	For a balanced answer that arrives at an analytically well substantiated judgement on the policy that would cause the least difficulty for the economy.	5
E2	For an answer that has balance and makes some attempt at a substantiated judgement on the policy that would cause the least difficulty for the economy.	3 – 4
E1	For an answer that has balance (i.e. considers the unintended consequences proposed), but with no convincing overall judgement that considers the different policies holistically.	1 – 2

- 5 In early 2022, Sri Lankans started experiencing power cuts and shortages of basics such as fuel. The rate of inflation rose to 50 per cent while unemployment rate remains high. Currently, Sri Lanka owes about \$7 billion to China and around \$1 billion to India.**

Adapted from BBC, 29 March 2023

- (a) Explain two possible causes of inflation. [10]
- (b) Discuss the extent to which the Sri Lankan government should focus on reducing its inflation rate over reducing its unemployment rate. [15]

Part (a) Question Interpretation

Command word/phrase	Explain	To utilise economic reasons and reasoning to explain the two possible reasons for inflation
Content	Causes of inflation	Demand-pull inflation → caused by increases in components of AD Cost-push inflation → caused by decreases in AS (increase in unit cost of production)
Context	No context	No specified context, students can utilize the examples given in the pre-amble or use their own examples

A relevant answer requires the students to explain the two causes of inflation (demand pull inflation caused by increases in any of the components of AD) and also cost-push inflation caused by decreases in AS due to increase in unit cost of production.

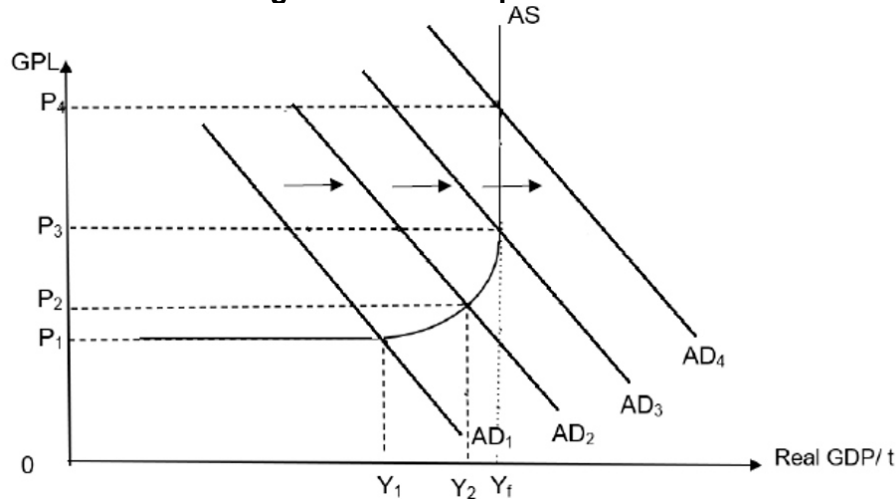
Introduction

- Inflation is a sustained increase in the general price level (GPL) of goods and services in the economy.
- There are two main types of inflation – demand-pull inflation and cost-push inflation. This response aims to explain the two causes of inflation.

Key Argument 1: Demand-pull inflation is caused by a rise in AD and it could be due to improved global sentiments.

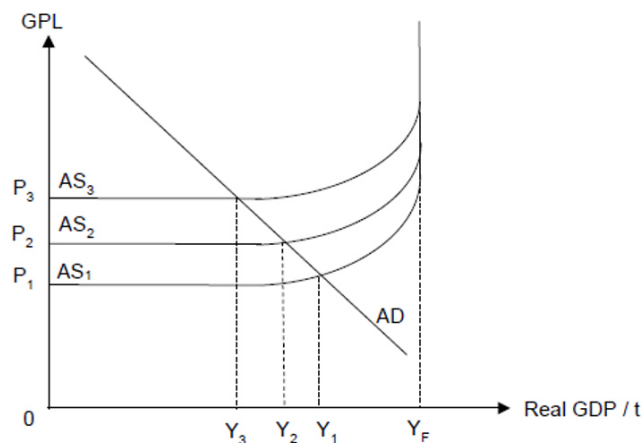
- When there is improved global sentiments, AD of the country will rise, causing demand-pull inflation.**
 - With improved global sentiments, firms are optimistic about the future state of the economy and their expected rate of return on investment would be higher, thus increasing investment expenditure (I).
 - At the same time, households around the world are more willing to spend on goods and services and therefore the net export revenue (X-M) of this country in question may increase due to higher export demand from foreign countries, assuming import expenditure remains constant.
 - The economy is initially in equilibrium at GPL of P1 and national output of Y1, as seen in the Figure 1 below.
 - This rise in investment expenditure and export revenue will lead to a rise in AD from AD1 to AD2. As AD rises from AD1 to AD2, the general price level (GPL) increases from P1 to P2 and real output level increases to Yf.
 - At full employment output of Yf, there is no more spare resources and real output cannot rise any further.
 - When AD continues to rise from AD2 to AD3, this simply causes firms to bid up factor prices as they compete for resources with one another.

- As there is no overall increase in utilisation of resources, but merely a transfer of resources from one firm to another, total output remains unchanged. The higher unit cost of production incurred by firms are then passed on to households in the form of higher prices of goods and services, resulting in the GPL escalating to P3.
- This increase in the GPL from P1 to P3 thus represents demand-pull inflation.

Figure 1: demand-pull inflation

Key Argument 2: Cost-push inflation is caused by a decrease in AS due to higher unit cost of production.

- **When there is an increase in the price of imported factor inputs, it will increase the unit cost of production, thus causing cost-push inflation.**
 - Imported inputs are key inputs to produce a variety of products, e.g., crude oil for non-oil producing countries.
 - An increase in price of imported crude oil drives up the price of energy, and energy is a necessary input in any production process.
 - The resulting rise in unit cost of production pushes up the prices of almost all products as reflected by increases in the GPL since the AS curve would be shifting up as AS falls.
 - In Figure 2 below, at the initial equilibrium where AD1 intersects with AS1, the initial general price level is P1 and the corresponding real output is Y1. When there is an increase in the unit cost of production at all levels of output, it reduces the profit margins of firms.
 - The short-run aggregate supply decreases and the AS curve shifts upwards from AS1 to AS2. Therefore, the general price level rises from P1 to P2 while real output falls from Y1 to Y2.
 - When the unit cost of production rises persistently, it leads to cost-push inflation.

Figure 2: Cost-push inflation

Mark scheme

Level	Knowledge, Application/Understanding and Analysis	Marks
L3	For a well-developed answer that has: ● good scope – explain the two causes of inflation (both demand-pull and cost-push), and; ● good rigour – clearly utilises appropriate AD/AS analysis to explain; and; ● good application and good use of appropriate diagrams.	8 – 10
L2	For an answer that: ● lacks scope – only explain either demand-pull or cost-push inflation; ● lacks rigour – there are gaps in explaining the causes of inflation; ● lacks application and lacks good use of appropriate diagrams to explain.	5 – 7
L1	For an undeveloped answer that: ● is descriptive, lacking in application of economic theory; and/or ● contains serious and pervasive conceptual errors; and/or ● is largely irrelevant.	1 – 4

- 5 In early 2022, Sri Lankans started experiencing power cuts and shortages of basics such as fuel. The rate of inflation rose to 50 per cent while unemployment rate remains high. Currently, Sri Lanka owes about \$7 billion to China and around \$1 billion to India.

Adapted from BBC, 29 March 2023

- (b) Discuss the extent to which the Sri Lankan government should focus on reducing its inflation rate over reducing its unemployment rate. [15]

Part (b) Question Interpretation

Command word/phrase	<i>Discuss the extent</i>	Provide multiple perspectives and come to a reasoned conclusion on the extent to which the Sri Lankan government should prioritise price stability or full employment
Content	<i>Reducing inflation rate..... unemployment rate... focus on</i>	Negative consequences of high inflation (goal of <i>price stability</i>) and negative consequences of high unemployment (goal of <i>full employment</i>)
Context	<i>Sri Lankan government</i>	Context of Sri Lanka

A relevant answer requires candidates to analyse the costs / negative consequences of high inflation and high unemployment on Sri Lanka. The evaluation entails a reasoned judgement on which negative consequence is worse, thus justifying the government focusing its attention on that particular goal.

Introduction

- As mentioned in the preamble, Sri Lanka is facing the twin issues of high inflation and high unemployment.
- Both macroeconomic problems lead to negative consequences on the economy and on Sri Lankan citizens' standard of living (SOL).

- As such, the Sri Lankan government should focus on resolving these issues. However, which macroeconomic issue the Sri Lankan government should focus on would depend on the relative severity of the negative consequences.

KA1: Sri Lankan government should focus on reducing inflation rate due to the negative consequences of high inflation.

- **High inflation leads to the negative consequences of negative growth and rise in unemployment.**
- With inflation, the real value of money falls, causing consumers to have a lower purchasing power. Therefore, households whose nominal income do not rise as fast as the general price level will suffer a fall in real income. They are able to consume fewer goods and services than before, and hence their material standard of living may worsen.
 - This could be particularly severe for Sri Lankans as the increase in inflation by 50% means that their material standard of living would have decreased significantly.
- Inflation erodes the real value of money, savers lose out because over time, their savings can buy fewer goods and services. Moreover, when prices are rising rapidly, rather than saving, consumers may prefer to spend more because they expect goods and services to be even more expensive in the future. With less savings, future consumption and future material standard of living may be lowered.
- Furthermore, when inflation is high and rising, investment will be discouraged. High levels of inflation increase the uncertainty of investment decisions, making it difficult for firms to predict their costs and revenues since inflation distorts price signals i.e., expected rate of return on investment is uncertain and volatile.
- Thus, investment will be discouraged, leading to a fall in AD. This triggers the reverse multiplier effect, leading to successive decreases in income-induced consumption, and a larger decrease in AD and hence multiplied fall in real GDP, causing negative growth.
- At the same time, as production level decreases, the derived demand for labour falls, increasing demand-deficient unemployment.

KA2: Sri Lankan government should focus on reducing unemployment rate due to the negative consequences of high unemployment.

- **High unemployment results in the negative consequences of negative actual and potential growth.**
- If the unemployment rate was high, as is the case in Sri Lanka (based on the preamble), it would mean that there are more households without a job, and hence no income.
 - This would mean lower MSOL as the purchasing power of households are lowered, leading to lower consumption of goods and services.
- Moreover, high unemployment that is prolonged leads to worries about possible loss of income sources, even amongst those currently employed.
 - This causes increased levels of anxiety for residents in the country about job security, lowering non-material living standards.
 - When unemployment is high and rising and/or prolonged, it leads to higher crime rates because of the lower opportunity cost of crime, and residents' sense of peace of mind is diminished which lowers non-material living standards (NMSOL).
- High unemployment therefore causes both MSOL and NMSOL of residents Sri Lanka to be lower.
- The high unemployment rate in Sri Lanka may lead to poor business confidence, causing fall in investments by firms as the expected returns of investments are likely to be low. This is especially so if the high unemployment is prolonged and/or rising.
 - Since investment is a component of the AD, AD will fall, creating a reverse multiplier effect on the country's real GDP, as explained previously. This will then result in negative growth, reduction in production and cause a fall in the derived demand for labour, further worsening the already high unemployment situation.

- Additionally, the fall in investment can also slow down potential growth since a fall in investment will cause the country's capital stock to expand more slowly, assuming that the growth rate of capital stock still exceeds the rate of capital depreciation.
- Hence, rising unemployment would have a negative effect on both actual and potential growth, as well as employment levels.

Evaluative Conclusion (a reasoned stand that follows from the above analysis + 2 well-developed ATMS evaluative angles):

- **[Stand + Situation]** Whether the Sri Lankan government should focus on reducing its inflation rate over reducing its unemployment rate would depend on which would bring about greater benefits for the economy in terms of reducing the negative effects associated with either inflation or unemployment. Given that Sri Lanka is facing an inflation rate of over 50%, which is a very high rate of inflation, Sri Lanka should focus on reducing the inflation rate first.
- **[Magnitude/Situation]** Given how a reduction in the inflation rate would have a positive effect by reducing unemployment rate, it would result in a significant improvement in the SOL of the Sri Lankans as we would see a rise in real GDP and thus, a rise in employment levels.
 - Also, given how the Sri Lankans do not have access to basic necessities such as power and fuel, a reduction in inflation would definitely bring about a greater SOL for the Sri Lankans, as it would make such necessities more affordable for the people.
 - With a reduction of high inflation rate, there would be restoration of confidence in the economy. This increased confidence in the economy would cause other countries to be still confident in lending Sri Lanka money to carry out its expansionary policies, as well as attracting more FDI to generate more jobs in Sri Lanka.

Mark scheme

Level	Knowledge, Application/Understanding and Analysis	Marks
L3	For a well-developed answer that has: • good scope and balance – explains the negative consequences of high inflation and high unemployment; and • good rigour – clearly explains the consequences with references to AD/AS analysis and standards of living; and • good application to the context of Sri Lanka.	8 – 10
L2	For an answer that: • lacks scope and balance – only explains the negative consequences of high inflation or high unemployment; • lacks rigour – there are gaps in explaining the negative consequences, without linking to AD/AS analysis or standard of living; • lacks application to the context of Sri Lanka or the pre-amble.	5 – 7
L1	For an undeveloped answer that: • is descriptive, lacking in application of economic theory; and/or • contains serious and pervasive conceptual errors; and/or • is largely irrelevant.	1 – 4
Level	Evaluation/Synthesis	Marks
E3	For a well-substantiated judgement on whether Sri Lankan government should focus on reducing high inflation or high unemployment.	5
E2	For an answer that makes some attempt at a substantiated judgement on whether Sri Lankan government should focus on reducing high inflation or high unemployment.	3 – 4
E1	For a reasoned evaluative comment on the strength of either the thesis or anti-thesis argument.	1 – 2

6 Deglobalisation is the movement towards a less connected world, characterised by powerful nation states, local solutions, and border controls, rather than global institutions, treaties, and free movement.

(a) Explain why a country like Singapore exports and imports both different and similar types of goods and services when it trades with other countries like China. [10]

Part a Question Interpretation

Command word/phrase	<i>Explain why</i>	To utilise economic reasons and reasoning to explain why Singapore exports and imports both different and similar types of goods in trade with China
Content	<i>Exports and imports different and similar types of goods with trade with China</i>	Patterns of Trade – inter-industry trade and intra-industry trade
Context	<i>Trade with China</i>	Context of Singapore, her trade with China

A relevant answer requires candidates to explain the reasons for inter-industry trade and intra-industry trade between Singapore and China.

Introduction

- International trade refers to the exchange of goods and services between countries. It can either be inter-industry or intra-industry trade.
- The trade pattern of a country refers to the composition and volume of its exports and imports, destination of its exports and the source of its imports.
- The trade pattern of inter-industry trade between Singapore and China can be explained by their theory of comparative advantage and the trade pattern of intra-industry trade can be explained by consumers' desire for variety and rational decision making.
- **Key Argument 1: Singapore exports and imports different goods with trade with China (inter-industry trade) because of the theory of comparative advantage.**
- A country has comparative advantage in producing a good when it can produce a good at lower opportunity cost than another country which entails sacrificing fewer units of another good in producing that good.
- A country tends to specialise and export the good for which it has CA in and import the good for which it has comparative disadvantage.
 - This is because in so doing, and assuming the terms of trade are mutually beneficial where the terms of trade lie between the 2 countries' opportunity cost, the trade partners enjoy a rise in consumption levels where consumption is at a level that lies beyond its PPC, leading to higher material living standards.
- The factor that determines a country's CA is its factor endowment and extent of capital accumulation
- Singapore has a relative abundance of physical and human capital, compared to land and (quantity of) labour, because of both domestic and foreign direct investment (FDI) in physical capital and its investment in human capital.
 - On the other hand, as a small country, it has limited land and labour. China's factor endowment is the opposite.
- As such if Singapore were to produce her own agricultural products, the opportunity cost, which entails forgoing the production of integrated circuits, will be high as its abundance of capital and high skilled labour force would have enable it to produce quite a lot of integrated circuits.

- In contrast, due to China's relative abundance of land and labour, its opportunity cost of producing agricultural products will be low and lower than Singapore because its lack of factors needed for producing integrated circuits would have limited the amount of this good that it could have produced, given its overall resources, when it devotes its resources to agricultural production.
- This means Singapore has a comparative disadvantage in agricultural production while China has the comparative advantage in this sector. On the other hand, Singapore has the comparative advantage in integrated circuits production and China has the comparative disadvantage in this sector.
- So long as the no. of units of agricultural product obtained in exchange for 1 unit of integrated circuit > opportunity cost of producing 1 integrated circuit, SG will be willing to export integrated circuit to China. So long as 1 unit of agricultural product imported from China < SG's opportunity cost of producing 1 agricultural product, SG will import the latter from China.
 - This explains why there could be inter-industry trade between Singapore and China.
- **Key Argument 2: Singapore exports and imports similar goods with trade with China (intra-industry trade) because of consumers' desire for variety of goods and rational decision making.**
- From Singapore consumers' point of view, so long as the Marginal Benefit obtained from buying instant noodles from China is > than the Marginal Cost (which is the price charged by the China producer), Singapore residents will be willing to import the good/service from China as it leads to rise in net benefit.
- China is likely to have the CA in producing instant noodles given the difference in factor endowment. China-made instant noodles are likely to be cheaper than SG-made ones if the instant noodles produced by both countries are assumed to be homogeneous. China producers are prepared to export instant noodles to China so long as the prices SG residents are willing to pay > their MC of production.
- However, Singapore also exports instant noodles to China. From SG producers' point of view, so long as China's residents are willing to pay a price (MR) for the SG-made instant noodles that > the MC of production, production is profitable, and SG producers are willing to export the good/service to China.
- The price charged by SG firms for instant noodles could be higher than the price charged by China firms since SG has comparative disadvantage in this good.
- However, the China residents might be willing to pay the higher price because their $MB \geq MC$ (price charged by SG producer) and this could be due to their desire for variety. The SG instant noodle firm that is producing a similar but differentiated good/service – e.g. the flavours or textures of the SG-made ones are different from the China made ones.
- Thus, horizontal intra-industry trade occurs between SG and China due to their residents' desire for variety.

Mark scheme

Level	Knowledge, Application/Understanding and Analysis	Marks
L3	For a well-developed answer that has: ● good scope – explain clearly the reasons for inter and intra industry trade; ● good rigour – explains using theory of comparative advantage and appropriate analysis; and ● good application to context of Singapore and China.	8 – 10
L2	For an answer that: ● lacks scope – only explain either the reasons for inter industry or intra industry trade; ● lacks rigour – gaps in explaining the above, and/or explanation is not supported by appropriate analysis; and/or ● lacks application to the context of Singapore and China.	5 – 7
L1	For an undeveloped answer that: ● is descriptive, lacking in application of economic theory; and/or ● contains serious and pervasive conceptual errors; and/or ● is largely irrelevant.	1 – 4

6 Deglobalisation is the movement towards a less connected world, characterised by powerful nation states, local solutions, and border controls, rather than global institutions, treaties, and free movement.

(b) With the trend of rising deglobalisation, discuss whether Singapore should also deglobalise. [15]

Part b Question Interpretation

Command word/phrase	<i>Discuss whether...</i>	To examine and present the different aspects of a problem or subject, and give a judgement on whether Singapore should deglobalise.
Content	<i>Deglobalise</i>	Deglobalise – movement towards less connected world, its benefits and costs
Context	<i>Singapore</i>	Context of Singapore economy

A relevant response requires candidates to explain the benefits and costs of deglobalization on Singapore's economy. The evaluation entails a judgement on whether Singapore should deglobalise.

Introduction

- Deglobalisation occurs when countries limit international trade (via protectionist measures such as import tariffs or quotas), labour flows (e.g. restrict the inflow of foreign labour) and/or impose capital controls.
- To decide whether Singapore should deglobalise, the disadvantages and advantages of deglobalisation should be considered in relation to the context of Singapore, a very small economy, and against the backdrop of the rising trend of deglobalisation.

KA1: Singapore should deglobalise because it has benefits to Singapore's economy

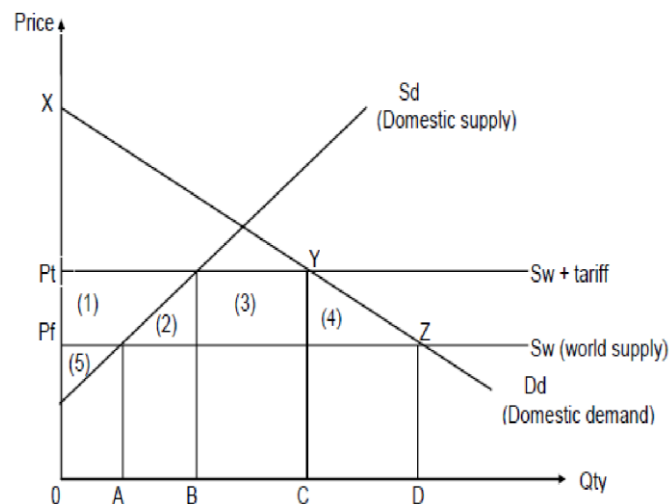
- **Deglobalisation leads to the benefits of reducing structural unemployment and reducing the high income inequality caused by globalisation.**
- When there is free movement of goods/services, countries will experience structural changes. The country will import goods/services in which it has comparative disadvantage because residents will choose to buy the relatively cheaper imported substitute and lower their demand for the domestic good. The domestic firms selling the substitutes of the imports face a fall in demand and will in turn lower their demand for labour.
 - This leads to structural unemployment due to the occupational immobility of workers. While globalisation enables the exporting sectors with CA to expand, which leads to job creation, those who lost their jobs in the import competing sectors may not have the relevant skills to take up the job openings in the exporting sectors. This results in structural unemployment.
- Free movement of capital through globalisation can also lead to structural unemployment.
 - For example, in a bid to cut costs, US MNCs have been off shoring some aspects of their production processes to developing countries like China / India where labour costs are relatively cheaper. The resulting closure of plants in USA led to structural unemployment in USA.
- Free trade and free capital movements also results in higher income inequality.
 - While wage rates will be falling in the declining sectors and structural unemployment rises, wage rates will be rising in the exporting sectors which are expanding, and in sectors receiving FDI. because demand for labour in these sectors will be rising. This leads to widening income gaps in the country.
- To avoid these problems of globalisation, a country, including SG might thus opt to deglobalise, just like what other countries have done, by imposing tariffs / quotas to limit imports or not allow FDI. This prevents the import-competing firms from declining, thus reducing structural unemployment and also reducing the high income inequality that resulted from globalisation.

(*Optional addition to KA1) Singapore should deglobalise to reduce her vulnerability to external shocks

- When a country is open to international trade, its growth performance and GPL becomes dependent on external factors.
 - For example, the US financial crisis of 2008 which led to a recession in USA caused USA's trade partners to also go into recession due to the falling demand by USA for its trade partners' exports when USA's GDP contracted.
 - For a country like SG for which X is a big share of its AD, the falling X leads to a significant fall in AD which leads to fall in real GDP (i.e. -ve growth) and rising demand-deficient unemployment.
- When the on-going Russian-Ukraine war broke out, the resulting shortage of grains led to imported inflation in countries like Singapore that depended on imported raw materials. The rise in prices of imported raw materials drove up the unit cost of production of firms in SG. This led to a fall in AS and hence rise in the general price level. Continuous rise in the prices of imported raw materials led to imported cost-push inflation.
- Deglobalisation by limiting dependence on exports and imported inputs can thus lower SG's exposure to external shocks.

KA2: Singapore should not deglobalise because it has costs to Singapore's economy.

- **Deglobalisation will lead to welfare loss to society as a whole.**
- A country like SG, a small country, is a price taker in world markets. Assume that for a certain good, foreign producers have the CA and the price of the imported good (P_f) is cheaper compared to if it were to be locally produced.
- If the government imposes an import tariff, it will cause the SS of the imported good to fall. In the diagram below, the SS falls and shifts up by the amount of tariff per unit.



- With the rise in price of the imported good, consumers lower the quantity demanded for the imported good and substitute towards the local good that is now relatively cheaper. The quantity of imports falls from AD to BC units.
- Overall, consumers lower Q_d for this good from D to C units and lose consumer surplus amounting to Area 1+2+3+4.
- Domestic firms are now able to charge a higher price (P_t) and increased Q_s from A to B units. This leads to a rise in producers' surplus by area 1.
- The govt earns area 3 as tariff revenue.
- There is a net loss of welfare to society of areas 2 and 4.
- Thus, although firms (and workers in this industry gain because there will be a fall in structural unemployment) gain, collectively, residents are worse off.

(*Optional addition to KA2) Singapore should not deglobalise as it will lead to higher cost of living, lower material living standards

- Singapore has hardly any natural resources. Without international trade, the cost of living will be much higher because if Singapore were to produce her own food crops and livestock, given the high demand relative to domestic supply, where domestic supply is low due to high MC from limited land, prices of food crops and livestock will be very high. This drives up the cost of living.
- Moreover, if Singapore were to lower exports, the reduced scale of production by domestic firms will lead to less exploitation of economies of scale (internal and external). As such, AC and MC of production rises, which translates to higher prices of goods and services.
- The inflow of foreign labour also helps to keep the cost of construction and various social services (e.g. cleaning, health care) lower as the inflow of such labour increases the supply of labour in these sectors and hence lowers wage rates. Deglobalisation in the form of limiting the inflow of foreign labour will lead to higher wage rates and higher MC of production faced by firms in these sectors, leading to higher prices and hence higher cost of living for residents.
- With higher cost of living, residents' purchasing power declines (assuming income remains constant), therefore resulting in lower material living standards.

Evaluative Conclusion (a reasoned stand that follows from the above analysis + 2 well-developed ATMS evaluative angles):

- **[Stand]** Singapore should not deglobalise because the advantages are likely to be outweighed by the disadvantages of doing so.
- **[Situation + Alternative]** Singapore is a very small and open economy. If she does not make use of exports and FDI to drive her growth, the alternative is to rely on domestic consumption and investment to drive growth.

- However, the local market is small given her small population. Growth from a rise in domestic demand will not be able to offset the negative growth due to reduction in external demand and FDI.
- Continued growth will also be limited because her resources are insufficient to accommodate the rise in AD caused by rise in domestic C and I. While it could count on technological advancement to create the potential growth to accommodate a rise in AD, the rate of technological advancement will likely be slow due to reduction in competition and absence of FDI.
- Hence, deglobalisation will result in a fall in real GDP initially and very limited rise in GDP over time. This adversely affects Singapore's material living standards. As such, even in a backdrop of rising deglobalisation, Singapore should continue to remain open and forge FTAs with countries that are not deglobalising, so that exports and FDI can continue to be a driver of economic growth.
- **[Alternatives]** Moreover, while globalisation has led to structural unemployment and higher income inequality, this can be mitigated by SS-side policies like provision of training opportunities and provision of incentives for workers to re-skill, so that they can take up jobs found in the sunrise industries that are thriving.
 - Not all countries are deglobalising and there is still growth potential for the export sector. As re-skilling takes time, the government could provide transfer payments in the shorter term to assist the structurally unemployed.
 - In addition, while globalisation exposes Singapore to imported inflation, cost of living in a deglobalised Singapore will likely to be higher than a globalised Singapore suffering imported inflation. This is due to her land scarcity. There are also ways to mitigate the effects of imported inflation – e.g. by having a strong SGD and by diversifying the sources of our imported inputs.

Mark scheme

Level	Knowledge, Application/Understanding and Analysis	Marks
L3	For a well-developed answer that has: • good scope and balance – analyses both the benefits and costs of deglobalization; and • good rigour – clearly explains the benefits and costs using appropriate analysis; and • good application to context of Singapore's economy.	8 – 10
L2	For an answer that: • lacks scope and balance – only explains either the benefits or costs of deglobalization; • lacks rigour – there are gaps in explaining the benefits and costs without appropriate analysis, and/or explanation is not supported by well-labelled and accurate diagram(s).	5 – 7
L1	For an undeveloped answer that: • is descriptive, lacking in application of economic theory; and/or • contains serious and pervasive conceptual errors; and/or • is largely irrelevant.	1 – 4
Level	Evaluation/Synthesis	Marks
E3	For a well-substantiated judgement on whether Singapore should deglobalise.	5
E2	For an answer that makes some attempt at a substantiated judgement on whether Singapore should deglobalise.	3 – 4
E1	For a reasoned questioning of either the thesis or anti-thesis argument.	1 – 2